

Decomposition, Isolation, and Quality Control Check of Multi-Vitamin Organic Components

CHE-243 Reverse Engineering Laboratory Report

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Multi-Vitamins are a crucial diet supplement for many, because of their numerous health benefits, appealing edible nature, and flavorful fruit or candy taste. This experiment focused on the decomposition, isolation, and analysis of the organic Vitamin-C content in a chewable multi-vitamin gummy. The Organic components were extracted from the aqueous, and isolated from the separation ingredients via separatory funnel extraction and Rotating Evaporator. The results were obtained from the High Performance Liquid Chromatography (HPLC) instrument, and compared to a sample of pure Vitamin-C organic components. This pure sample was obtained through repeating the exact procedure for the multi-Vitamin sample. Both of the peaks containing the Vitamin-C, in the form of Ascorbic acid, matched identically, reflecting the proper amount of Vitamin-C present.

Introduction

Vitamin C is one of the most useful water-soluble vitamins, because of its many benefits. The major properties include anti-oxidation, cell repair and growth, and increased capacities to store iron. Vitamin C, commonly known as ascorbic acid, is not made by the body but is found in many food sources. A well-rounded diet includes the proper amount of Vitamin C a regular human body needs to function properly. However, for those who do not consume a well-rounded diet, it is recommended to take a multi-vitamin supplement that includes Vitamin C. The objective was to take a chewable multi-vitamin gummy, extract the organic Vitamin content, and isolate the Ascorbic acid. The isolated Ascorbic acid could then be compared with a pure sample of Ascorbic acid. Both samples would be dissolved in a methanol solution for analysis. The goal was to determine if the multi-vitamin sample contained the desired amount of Vitamin that was claimed, and to compare it to the pure vitamin sample.

Sections: Theory, Experimental Procedure, Results, Conclusion

This report has been divided into 4 section containing the Theory behind the work; the laboratory Work that was performed; the results obtained; and the final conclusions.

Before any experimental work can be performed, it is important

to recognize that there may be no results obtained. Time is limited, the resources are limited, and each project must follow strict guidelines to ensure that all students are granted access to the instruments. With these limitations in place, data was obtained after the completion of the laboratory work.

1. Theory

A chewable multi-vitamin gummy can be dissolved in water, however this presents a challenge of what to do with the dissolved impurities, and more importantly, the dissolved gelatin. The gelatin cannot be run in any of the instruments, as it is too thick and would clog the instruments before they are able to produce useful data. Dissolving a multi-vitamin gummy in a 50:50 water methanol solution is the proper way to ensure that the gummy fully dissolves. The gelatin, and other impurities, are insoluble in methanol, thus using a solution that contains equal parts water and methanol will retain both aspects of the gummy. Once the gummy is dissolved, diethyl ether and saturated sodium chloride can be added to form an organic layer, and push all organic contents out of the aqueous layer. The aqueous is extracted, and the organic layer is prepared in the Rotating Evaporator Instrument. Once completed, the resulting pure vitamin components can be separated by cooling the methanol solution to 0°C. Concentrated Hydrochloric acid is added to lower the pH, which promotes the recrystallization of Vitamin-C in the stable form of L-Ascorbic Acid, but not the other B Vitamins.

2. Experimental Procedure

Beginning on the first lab session, 25ml of methanol was added to a 50ml beaker. 25 ml of water was added to the same beaker, and the solution was swirled to mix. The multi-vitamin gummy was added to the solution, and the initial temperature was recorded to be 25°C. The beaker was added to a steam bath, and the gummy started to dissolve at 29°C. After the temperature reached 44°C, the gummy was almost completely dissolved, but still had a small amount left. The gummy completely dissolved at 50°C, and the beaker was removed at 51.5°C. The solution was allowed to cool, then 5ml. of concentrated sulfuric acid was added. The sulfuric

acid lowered the pH to the 0 to 1 range. The acidity was confirmed with red and blue litmus paper. The solution was added to an ice bath, and ice was added to the solution to further lower the temperature until it reached 3°C, then the contents were transferred to a 125ml Erlenmeyer flask. Once the solution was at -5°C, it was removed from the ice bath, and the solution was started to drain in a Buchner funnel. The crystals were too small, and the dissolved gelatin clogged the filter paper. This resulted in no crystals being collected. This marked the end of Trial 1, discussed more in Results.

To confirm that gelatin was only soluble in water, 20ml of methanol was added to a 50ml beaker, and a gummy was added. The solution was placed over the steam bath. At 46°C, the gummy was dissolving, but the gelatin remained as a sticky mess. At 55°C, right before the methanol started boiling, the gummy had reduced to about two-thirds its original size, and it was extremely sticky, making a mess at the bottom of the flask. Water was added, and the entire gummy completely dissolved.

The second lab session was started in a similar way. 25ml of methanol was added to a 125ml Erlenmeyer flask, and 25ml of water was added. The gummy was added to the flask, and the solution was placed over the steam bath. After 15 minutes, almost all of the gummy had dissolved, and the temperature had reached 51°C. The gummy completely dissolved at 56°C and was removed from the steam bath. The solution was added to a 125ml Separatory funnel and 15ml of diethyl ether was added to the funnel. 10ml of saturated sodium chloride was added to the funnel. The funnel was sealed, and shaken well, while occasionally stopping to release the pressure. 10 more ml of saturated sodium chloride was added to the funnel to increase the organic layer, and the funnel was sealed, then shaken again. The aqueous layer was extracted, and the organic layer was transferred to a 50ml round bottom flask. 5ml of diethyl ether was used to rinse the funnel, and it was added to the flask. The round bottom flask was placed in the Rotating Evaporator, and was allowed to continue until the volume was constant, and no more bubbles were present in the flask. The flask was removed, and 10 ml of methanol was added to the solution to increase the volume. The flask was cooled, but hardly any crystals would be present, and the crystals that did appear were far too small to be useful. The flask was placed back into the rotating evaporator to try and remove all of the methanol, but this did not work, as the solution would not be able to proceed to the HPLC afterwards.

The flask was removed from the Rotating Evaporator, and 3 ml of HCl was added to promote crystallization of the vitamin-c. The pH was tested, and confirm to be acidic. The flask was cooled, but no crystals appeared.

Starting the third lab session, 25ml of water was measured than added to a 125ml Erlenmeyer flask, along with 25ml of methanol. The gummy was added, the the solution was placed over heat. A second sample was prepared using 25ml of water, and 25ml

of methanol in a 125ml Erlenmeyer flask. This second flask was labeled, and a Vitamin-C gummy was added to the solution. The flask was placed on heat. The multi-vitamin solution was heated to 64°C, when the entire gummy dissolved, and the solution became homogeneous. It was removed from heat. The pure Vitamin-C solution was heated to 62°C, and removed from the heat. The multi-vitamin was added to a seperatory funnel and 20ml of ditheyl ether was added to the funnel. 20ml of saturated Sodium Chloride was added to the funnel. The funnel was sealed then shook to push all organic contents out of the aqueous layer. The aqueous was extracted, then the organic was placed in a 50ml round bottom flask. The flask was placed into the Rotating Evaporator, and it was allowed to continue for 18 minutes before being removed.

The pure Vitamin-C solution followed the exact same steps as the multi-vitamin: organic layer created, funnel shook, aqueous extracted, organic removed and placed in Rotating Evaporator. The sample was allowed to continue for 13 minutes before being removed.

Both of the solutions were ready to run in the HPLC instrument, however, there was a wait for the instrument. The wait would have pushed well past the ending time of the session, resulting in both the samples being labeled, and placed in storage until the following week.

Both of the samples were now ready to be run in the HPLC. The pure Vitamin-C sample was run first, followed by the multi-vitamin. The graphs, and results are attached in the appropriate section.

3. Results (*Data Tables and Graphs*)

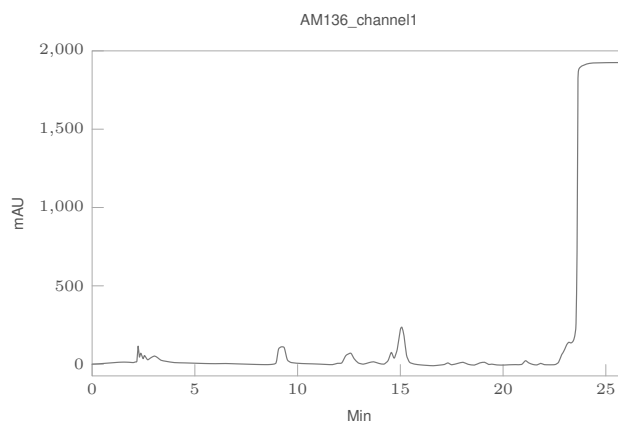


Fig. 1. Reconstructed chromatogram for the multivitamin sample, channel 1.

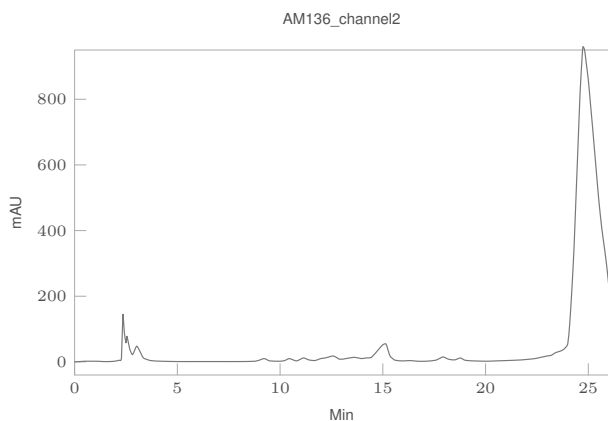


Fig. 2. Reconstructed chromatogram for the multivitamin sample, channel 2.

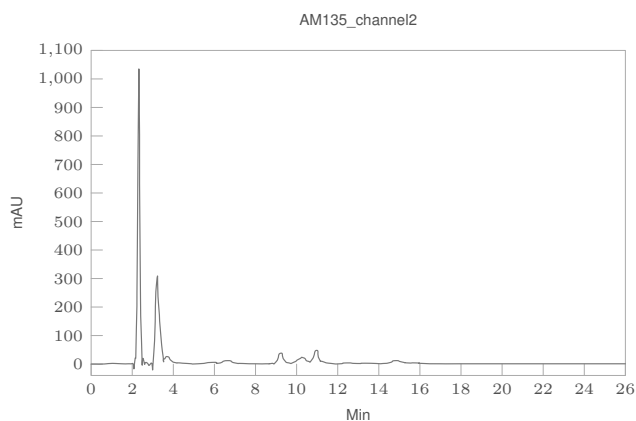


Fig. 3. Reconstructed chromatogram for the pure vitamin C sample, channel 2.

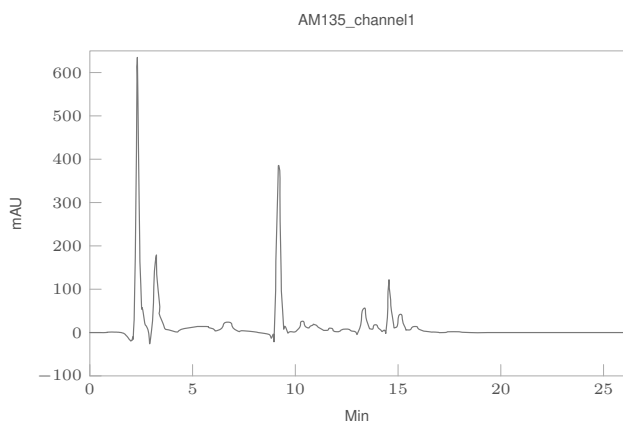


Fig. 4. Reconstructed chromatogram for the pure vitamin C sample, channel 1.

Table 1. AM136 channel1 peak results.

Idx	<i>t</i>	<i>Q</i>	<i>H</i>	<i>A</i>	<i>A</i> %
1	1.05	0.27	4.5	3.0	0.268
2	2.37	1.44	118.3	16.3	1.444
3	2.56	1.17	66.2	13.2	1.165
4	3.03	2.45	52.1	27.7	2.453
5	5.10	0.11	7.1	1.3	0.113
6	6.25	0.14	8.5	1.6	0.137
7	6.49	0.28	10.0	3.2	0.284
8	9.07	0.50	44.2	5.6	0.500
9	11.14	0.29	8.2	3.3	0.292
10	11.93	0.38	11.0	4.3	0.379
11	12.59	3.08	86.4	34.7	3.076
12	13.48	1.23	28.6	13.9	1.233
13	14.58	4.21	132.6	47.5	4.207
14	15.08	9.13	266.5	103.2	9.132
15	16.92	0.06	3.3	0.7	0.064
16	17.31	0.63	18.5	7.1	0.626
17	17.80	1.18	37.4	13.3	1.181
18	18.87	1.40	34.0	15.8	1.400
19	19.45	0.67	27.1	7.6	0.671
20	21.09	1.40	45.4	15.8	1.396
21	21.83	0.15	5.5	1.7	0.150
22	22.91	2.38	74.9	26.9	2.381
23	23.64	67.45	1402.2	761.9	67.446
Tot	—	100.00	2492.4	1129.7	100.000

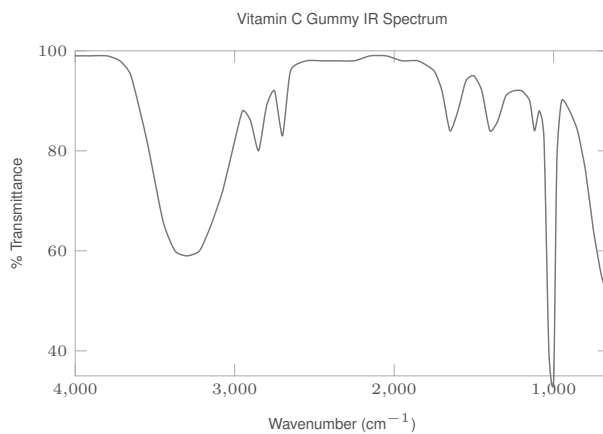


Fig. 5. Reconstructed IR spectrum, Pure Vitamin-C Gummy

Table 2. AM136 channel2 peak results.

Idx	<i>t</i>	<i>Q</i>	<i>H</i>	<i>A</i>	<i>A</i> %
1	0.16	0.00	0.3	0.0	0.002
2	2.35	1.77	147.1	20.6	1.765
3	2.55	1.41	74.7	16.4	1.407
4	3.02	1.76	46.9	20.6	1.763
5	5.09	0.02	1.7	0.3	0.023
6	9.10	0.14	9.0	1.6	0.138
7	9.31	0.24	10.1	2.8	0.243
8	10.22	0.04	1.6	0.5	0.045
9	11.14	0.61	15.8	7.1	0.612
10	11.97	0.06	1.9	0.7	0.057
11	12.59	0.72	21.2	8.5	0.724
12	13.62	0.12	3.7	1.4	0.124
13	14.17	0.06	2.9	0.7	0.061
14	14.82	0.45	25.4	5.3	0.453
15	15.16	1.38	38.7	16.2	1.385
16	16.32	0.02	1.3	0.3	0.023
17	17.57	0.01	1.2	0.2	0.014
18	17.93	0.55	17.5	6.4	0.546
19	18.77	0.51	12.5	5.9	0.507
20	20.07	0.10	3.7	1.1	0.096
21	21.74	0.09	2.0	1.0	0.087
22	24.74	89.93	738.0	1049.5	89.928
Tot	–	100.00	1177.4	1167.0	100.000

Table 3. AM135 channel2 peak results.

Idx	<i>t</i>	<i>Q</i>	<i>H</i>	<i>A</i>	<i>A</i> %
1	0.13	0.00	0.1	0.0	0.004
2	0.93	0.01	0.4	0.0	0.010
3	2.30	61.84	1059.3	218.9	61.837
4	3.18	23.15	280.4	81.9	23.151
5	3.55	0.44	17.5	1.5	0.438
6	3.72	2.18	25.2	7.7	2.178
7	4.32	0.10	2.4	0.4	0.101
8	5.92	0.27	4.1	1.0	0.270
9	6.63	0.90	12.4	3.2	0.898
10	8.76	0.03	0.9	0.1	0.034
11	9.22	2.56	38.3	9.1	2.560
12	9.63	0.14	2.5	0.5	0.137
13	10.13	1.01	15.6	3.6	1.013
14	10.33	1.32	21.9	4.7	1.320
15	10.97	3.90	46.5	13.8	3.898
16	11.71	0.04	0.9	0.2	0.042
17	12.43	0.25	4.0	0.9	0.247
18	12.94	0.07	1.2	0.3	0.071
19	13.33	0.22	3.4	0.8	0.216
20	13.88	0.11	2.2	0.4	0.106
21	14.53	0.25	5.3	0.9	0.253
22	14.80	0.69	11.6	2.4	0.691
23	15.14	0.32	5.7	1.1	0.317
24	15.82	0.20	4.2	0.7	0.205
Tot	–	100.00	1566.0	353.9	100.000

Table 4. AM135 channel1 peak results.

Idx	<i>t</i>	<i>Q</i>	<i>H</i>	<i>A</i>	<i>A</i> %
1	0.93	0.01	0.4	0.0	0.007
2	2.30	27.79	675.2	153.0	27.795
3	3.18	19.17	191.7	112.0	20.341
4	4.32	4.80	38.3	26.4	4.801
5	4.44	4.29	26.5	24.5	4.443
6	5.28	0.29	11.0	1.6	0.292
7	6.71	1.73	28.9	9.5	1.726
8	7.96	0.04	1.4	0.2	0.042
9	8.78	0.02	1.3	0.1	0.022
10	9.22	18.96	403.9	104.3	18.956
11	10.14	1.32	27.6	7.3	1.322
12	10.33	1.41	35.3	7.7	1.405
13	10.79	1.04	23.3	5.7	1.042
14	10.96	1.34	23.0	7.4	1.341
15	11.73	1.04	20.5	5.7	1.038
16	12.44	0.72	16.5	4.0	0.719
17	12.83	0.28	5.2	1.6	0.285
18	13.32	2.64	70.7	14.6	2.644
19	13.89	0.99	21.2	5.5	0.990
20	14.53	6.64	163.0	36.5	6.637
21	15.10	2.62	56.1	14.4	2.617
22	15.82	0.93	22.8	5.1	0.935
23	16.83	0.03	0.8	0.2	0.032
24	17.20	0.01	0.5	0.1	0.013
25	17.69	0.10	3.7	0.6	0.102
26	17.86	0.15	4.3	0.8	0.146
27	19.36	0.01	0.5	0.1	0.012
28	20.52	0.01	0.3	0.0	0.006
29	24.82	0.19	2.8	1.1	0.193
30	25.87	0.10	2.7	0.5	0.098
Tot	–	100.00	1879.7	550.5	100.000

Results Discussion

During the initial trial, the multi-vitamin was dissolved, the solution was cooled, and filtered using a Buchner funnel with vacuum. The initial error was not separating the aqueous components first, before vacuum filtering. The gelatin dissolved in the solution was too viscous for the filter paper, and clogged it. When the concentrated acid was added to lower the pH of the solution, it should have also slightly dissolved the remaining gelatin and other impurities. However, the acid did not promote recrystallization of the Vitamin C, and nothing else was able to be done with the solution, as it was now too acidic.

Trial one provided a great deal of success for future trials. The next trials now needed to have the aqueous removed, which would take the gelatin out of the solution, allowing the crystallized vitamin C to remain on the filter, and the solution to freely flow past. The second trial was performed following a similar approach. This time, when the gummy was fully dissolved, diethyl ether was added to promote the creation and separation of an organic layer. A polar solvent was added to push all of the organic components out of the aqueous, sodium chloride. The aqueous was extracted, and the organic was placed in the rotating evaporator to remove the diethyl ether.

The solution was cooled and the crystals that formed were too small for the filter paper available. This promoted a new idea: The Rotating Evaporator was running just below 60°C, and the B.P. for methanol was about 60°C. If the rotating Evaporator was turned up, it would remove all of the methanol, leaving behind a pure sample of organic vitamin contents. Once the process was started, it was then realized that this would leave behind a possibly solid organic, which could not be run in any instrument, as the strict requirements was a liquid, specifically for the HPLC. Additionally, there would be no way to separate the vitamins with this approach. Because of these reasons, the flask was removed from the instrument, and the trial was ended.

The third trial combined parts of all the previous. Instead of trying to separate the vitamins, this time, the organic contents were placed into the HPLC as they were. The same procedure had applied as before, except this time, a second flask was used to contain a pure vitamin C gummy. The gummies were dissolved with heat, the organic layer was created, and a polar solvent was added to push everything out. The rotating Evaporator was used, except this time, the temperature remained below 60°C, causing only the ether to be removed, and not the methanol. The resulting organic contents now contained the Vitamins, some impurities, and methanol. There was no proposed way to efficiently remove the methanol from the sample without losing some of the organic contents and retaining the ability to run the sample in the instruments. Both samples were run in the HPLC, and the pure vitamin C sample was placed in the IR. The graphs were obtained and constructed in the results section.

The IR spectroscopy graph of the pure vitamin C reflected a

mostly methanol signature, with a small fingerprint region that distinctly indicated the presence of ascorbic acid. One of the signature peaks was present for ascorbic acid at 2800-2900 cm⁻¹, one small peak at 1750, and the distinct peak of ascorbic acid at 1000 cm⁻¹. These peaks were small on the graph, compared to a pure ascorbic acid sample, however, this can be explained by most of the sample being methanol. The methanol part was influencing the graph more than the ascorbic acid.

Upon researching the appropriate HPLC signature of ascorbic acid, the most accepted peak appears at just past 2 minutes, and up to 4 minutes. Both graphs obtained from the multi-vitamin and the pure vitamin sample, contained distinct peaks at almost the same time, 2.3 minutes. This clearly reflects the ascorbic acid presence. The peaks for the multi-vitamin looked like the peaks for the pure sample, until the 22.5-minute mark. There was an unknown impurity that causes a sharp jump from the highest point of about 266 mAU, to about 1400 mAU. This unknown could have been any impurity present in the initial multi-vitamin gummy, but it could also be any remaining hydrophobic components. The remaining hydrophobic samples that do not run through the column completely can get stuck and cause the sharp increased peak at the end of the run, when they become unstuck from the column. This slightly skews the proportionality of the graph appearance and the area percent for the results, but for the final conclusion, the peak height and time at which it occurred was used.

Both results were obtained from channel 1 and 2, but for the analysis and conclusion, channel 1 is used. The time for the ascorbic acid peak in the multi-vitamin was 2.37, while the time for the pure vitamin C was 2.30. These two reflect the ascorbic acid presence, and the height can be compared. Because of unknown impurities, and dramatic increase at the end of the graph for the multi-vitamin, the area percent cannot be used to conclude anything valuable. Instead, the original claimed ascorbic acid content will be used. None of the tests that were performed could confirm the claim of the manufacturers. However, using the initial ascorbic acid content of the Multi-vitamin, and the original content for the pure vitamin C, a percent can be made, and the results from the HPLC can be analyzed.

Sample	Pure Vitamin-C	Multi-Vitamin
mg Ascorbic Acid	282 mg	75 mg
Time for Peak	2.30	2.37
% Area	27.79%	1.44%
Height	675.2	118.3

The multi-Vitamin claims to have 75mg of ascorbic acid present, and the pure vitamin C claims to have 282 mg. This results in a ratio of 75:282, or 27%. The HPLC peaks resulted in a height of 118.3 for the peak at 2.37 minutes for the multi-vitamin, and a peak at time 2.30 minutes, with height 675.2 for the pure vitamin C sample. Converting these two into ratios yields 17.5%. This

reflects a percent error of 34.5%, which relates to the quality of the multi-vitamins to not reflect accurately what they claim.

Regardless of the results obtained, it can be clearly seen that the ascorbic acid in each sample was close to the claimed amount. The area of the peaks for each was about half of the claimed milligrams of the ascorbic acid. Neither vitamin can be concluded to be more or less accurate, but the two can be observed to have an error of about 35%. This error can be explained by the many impurities, as the multi-vitamin numerous ingredients. However, despite the multi-vitamin containing more ingredients, the pure vitamin contained more unknowns, 26 compared to 23. This can be explained from the extraction phase. Most of the ingredients in the multi-vitamin were heavy, and not soluble in methanol. When the organic was extracted, these impurities had already left with the aqueous. Contrarily, the pure vitamin C gummy contains a small amount of ingredients, but they are mostly soluble in methanol, and not as heavy as the others.

Theoretical Peak %	$\frac{75mg}{282mg} = 27\%$
Calculated Peak %	$\frac{118.3}{675.2} = 17.5\%$
% Error	34.5%

4. Conclusion

The goal to decompose, isolate, and analyze the contents of a multi-vitamin, compared to a pure vitamin sample, was achieved successfully. Both initial gummies were dissolved, the organic contents were isolated and analyzed through the HPLC instrument. The next phase of this project would be determined the amount of ascorbic acid present in each of the samples. This would be done by either continuing with the recrystallization results and carefully tracking all weights to ensure no loss of contents, comparing the final weight of the crystallized ascorbic acid with the claimed weight or using the HPLC. A calibration curve would be made with pure vitamin C, obtained from a manufacturer of pure ascorbic acid, and not decomposed from a gummy. This would yield a very accurate result, as there would be no error from losing any of the solid when weighing the contents. Additionally, this can be expanded to analyze more Vitamins in the multi-vitamin gummy. Other interests included the B vitamins, zinc complexes, and flavoring added. These additional ideas require more time than the original, and will be left to the future operator of the laboratory to decide on performing them.